Linear Mixed Models (LMM)

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In this chapter we introduce linear mixed models. We start from simple designs in which the response is measured repeatedly on the same experimental unit and analyse them first with a **two-stage analysis**: we summarise the data of each experimental unit and analyse the summaries. The linear mixed model does both stages in one step. The last part briefly summarises the theory of Chapter 4 of Ruppert et al. (2003); the full derivations are in the extended version of this chapter.

```
library(tidyverse)
library(lme4)
library(Matrix)
library(QFeatures)
library(BiocFileCache)
```

Experimental and observational units

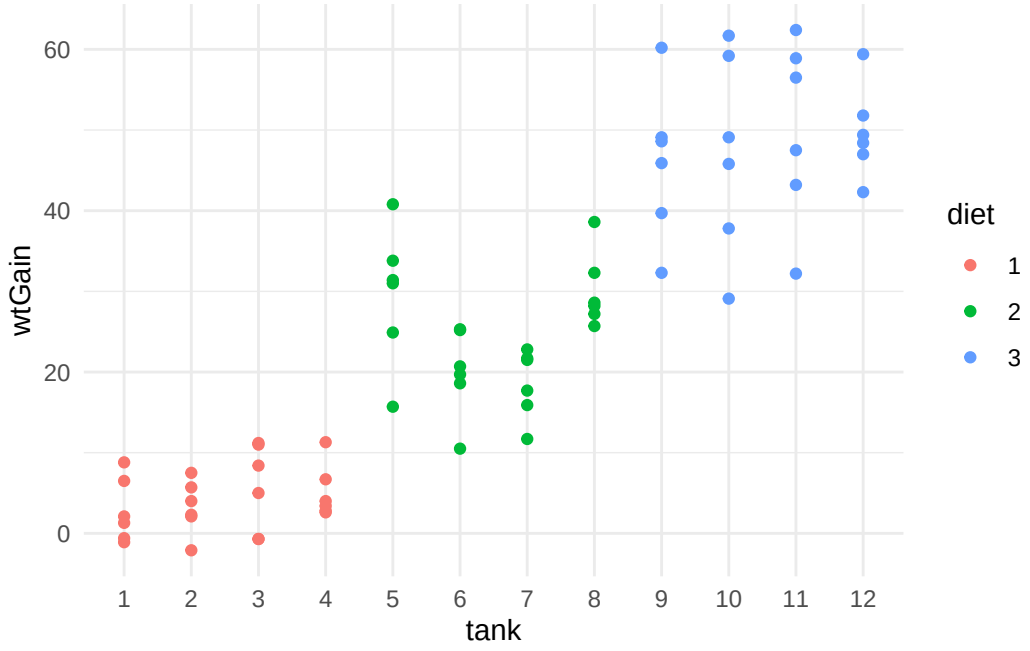
Whether we need random effects depends on the design: on which unit the treatment is randomised (the **experimental unit**) and on which unit the response is measured (the **observational unit**). We first illustrate this with a simple example and then with one protein from two proteomics experiments that are also used in the other lectures.

Pseudo-replication: fish in tanks

Researchers want to assess the effect of three diets on the weight gain of fish. They have 12 tanks with the same number of fish. Each diet is randomly assigned to 4 tanks, and at the end of the experiment the weight gain of 6 fish in each tank is recorded.

```r
fish <- data.frame(
  diet = factor(rep(1:3, each = 24)),
  tank = factor(rep(1:12, each = 6)),
  wtGain = c(
    8.8, 6.5, -1.1, 1.3, -0.6, 2.1,
    5.7, 4, -2.1, 2.3, 2.1, 7.5,
    -0.7, 11.2, -0.7, 11, 8.4, 5,
    3.4, 6.7, 11.3, 4, 2.6, 2.8,
    31, 15.7, 40.8, 33.8, 24.9, 31.4,
    10.5, 25.3, 18.6, 19.7, 20.7, 25.2,
    21.7, 15.9, 22.8, 21.5, 11.7, 17.7,
    38.6, 25.7, 32.3, 28.2, 28.6, 27.2,
    49.1, 39.7, 45.9, 48.6, 32.3, 60.2,
    61.7, 29.1, 49.1, 59.2, 45.8, 37.8,
    58.9, 43.2, 56.5, 47.5, 32.2, 62.4,
    59.4, 49.4, 42.3, 47, 48.4, 51.8))

ggplot(fish, aes(x = tank, y = wtGain, colour = diet)) +
  geom_point() +
  theme_minimal()
```



Naive data analysis:

```r
naiveFish <- lm(wtGain ~ diet, data = fish)
summary(naiveFish)$coef
```

```
             Estimate Std. Error   t value     Pr(>|t|)
(Intercept)  4.229167   1.528121  2.767560 7.244420e-03
diet2       20.333333   2.161090  9.408834 5.408316e-14
diet3       44.000000   2.161090 20.360099 6.933229e-31
```

Experimental units and observational units?

- The diet is randomised to the tanks: the 12 tanks are the experimental units.
- The weight gain is measured on the fish: the 72 fish are the observational units.
- Fish in the same tank share the same environment (water quality, temperature, feeding, …): they are **pseudo-replicates**. Weighing more fish of the same tank does not give more independent tanks.

Two-stage analysis

The diet effect has to be assessed with the variability between the experimental units, the tanks. A simple and correct analysis therefore works in two stages:

- **Stage I**: summarise the data of each experimental unit, here the average weight gain in each tank.
- **Stage II**: analyse the summaries with an ordinary linear model: 4 tanks per diet.

```r
stageIFish <- fish |>
  group_by(tank, diet) |>
  summarise(wtGain = mean(wtGain), .groups = "drop")
stageIFish
```

```
# A tibble: 12 x 3
   tank  diet  wtGain
   <fct> <fct>  <dbl>
 1 1     1       2.83
 2 2     1       3.25
 3 3     1       5.7
 4 4     1       5.13
 5 5     2      29.6
 6 6     2      20
 7 7     2      18.6
```

```
 8 8     2      30.1
 9 9     3      46.0
10 10    3      47.1
11 11    3      50.1
12 12    3      49.7
```

```
twoStageFish <- lm(wtGain ~ diet, data = stageIFish)
summary(twoStageFish)$coef
```

```
             Estimate Std. Error   t value     Pr(>|t|)
(Intercept)  4.229167   1.907782  2.216798 5.384469e-02
diet2       20.333333   2.698011  7.536416 3.554557e-05
diet3       44.000000   2.698011 16.308310 5.445651e-08
```

If we act as if the 72 fish are independent, the standard errors are too small: the analysis treats the 24 fish per diet as independent repeats, while there are only 4 independent tanks per diet.

Note that we cannot correct for the tanks with a fixed tank effect as in a block design: once we know the tank, we also know the diet. The tank effects are aliased with the diet effects.

```
coef(lm(wtGain ~ diet + tank, data = fish))
```

```
(Intercept)       diet2       diet3       tank2       tank3       tank4
  2.8333333  27.2666667  46.8833333   0.4166667   2.8666667   2.3000000
      tank5       tank6       tank7       tank8       tank9      tank10
 -0.5000000 -10.1000000 -11.5500000          NA  -3.7500000  -2.6000000
     tank11      tank12
  0.4000000          NA
```

Linear mixed model

The two-stage analysis is easy, but it has drawbacks:

- Information is lost when the data of a tank are summarised by a single number.
- It is only valid when all Stage I summaries are equally precise, i.e. when the design is balanced. With unequal numbers of fish per tank, or missing values, the precision of the tank averages differs, and Stage II does not account for that.

A **linear mixed model** combines both stages into one model:

$$\begin{cases} y_{tk} &= \beta_0 + \beta_{d(t)} + b_t + \epsilon_{tk} \\ b_t &\sim N(0, \sigma_b^2), \\ \epsilon_{tk} &\sim N(0, \sigma_\epsilon^2) \end{cases}$$

with y_{tk} the weight gain of fish k in tank t, $\beta_{d(t)}$ the effect of the diet of tank t and b_t a **random effect** for tank t. (Note, that the random effects and errors are assumed to be independent.)

- The diet effects are **fixed effects**: they are the parameters of interest.
- The tank effects are **random effects**: the 12 tanks are a random sample of tanks, and we are not interested in the effect of these particular tanks. They only cost one extra parameter, the **variance component** σ_b^2, the between-tank variance. σ_ϵ^2 is the within-tank variance.

```
lmmFish <- lmer(wtGain ~ diet + (1|tank), data = fish)
summary(lmmFish)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: wtGain ~ diet + (1 | tank)
   Data: fish

REML criterion at convergence: 481.9

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.60555 -0.45452 -0.05768  0.50128  1.97624

Random effects:
 Groups   Name        Variance Std.Dev.
 tank     (Intercept)  6.001   2.450
 Residual             51.348   7.166
Number of obs: 72, groups:  tank, 12

Fixed effects:
            Estimate Std. Error t value
(Intercept)    4.229      1.908   2.217
diet2         20.333      2.698   7.536
diet3         44.000      2.698  16.308
```

```
Correlation of Fixed Effects:
      (Intr) diet2
diet2 -0.707
diet3 -0.707  0.500
```

Comparisons between analyses

```
rbind(
  wrong = summary(naiveFish)$coef["diet2", 1:3],
  twoStage = summary(twoStageFish)$coef["diet2", 1:3],
  mixed = summary(lmmFish)$coef["diet2", 1:3])
```

```
         Estimate Std. Error  t value
wrong    20.33333   2.161090 9.408834
twoStage 20.33333   2.698011 7.536416
mixed    20.33333   2.698011 7.536416
```

The two-stage analysis and the mixed model give identical estimates, standard errors and t-statistics. For a balanced design with m tanks per diet and n fish per tank, the variance of a diet mean is

$$\text{var}\left[\bar{y}_d\right] = \frac{\sigma_b^2}{m} + \frac{\sigma_\epsilon^2}{mn}.$$

The between-tank variance is only averaged over the $m = 4$ independent tanks: weighing more fish per tank cannot reduce it. The mixed model accounts for both sources of variability.

Induced correlation

The random tank effect induces a correlation between fish of the same tank ($k \neq k'$):

$$\text{cov}\left[y_{tk}, y_{tk'}\right] = \text{var}\left[b_t\right] = \sigma_b^2, \qquad \text{corr}\left[y_{tk}, y_{tk'}\right] = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_\epsilon^2}.$$

Fish of different tanks remain independent. All fish of the same tank have the same variance $\sigma_b^2 + \sigma_\epsilon^2$ and the same correlation: this covariance structure is called **compound symmetry**. It is natural here, because the fish within a tank are exchangeable.

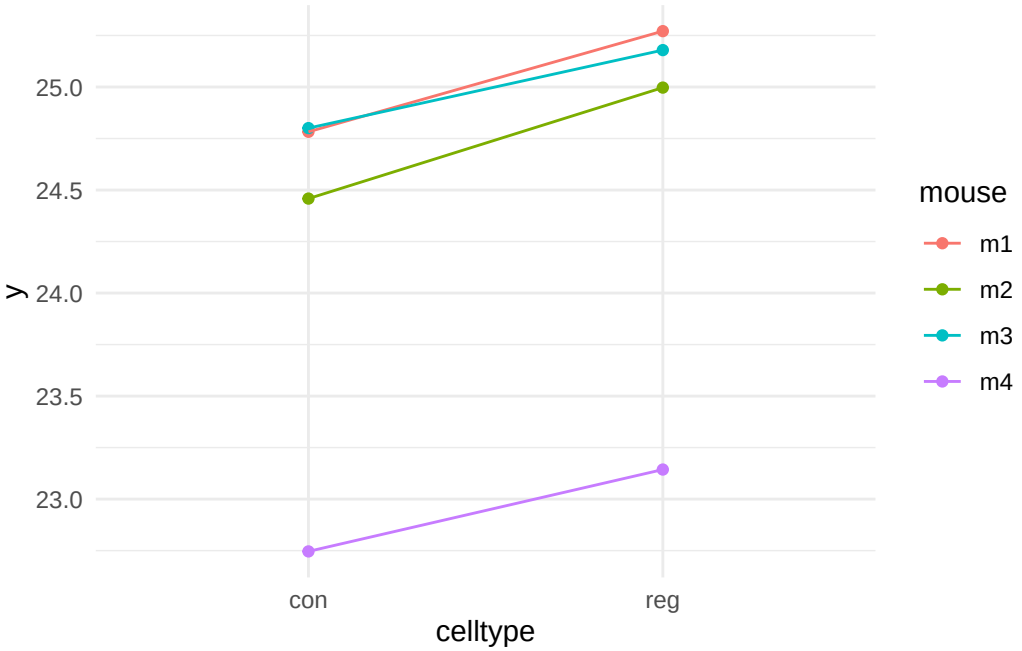
Blocking: mouse T-cells

In the mouse T-cell experiment, regulatory (Treg) and conventional (Tconv) T-cells are purified from each of 4 mice. Here, the cell type varies **within** each mouse: the mouse is a **block**, a randomised complete block (RCB) design.

```
bfc <- BiocFileCache()
mouse <- readRDS(bfcrpath(bfc,"https://github.com/statOmics/msqrob2data/raw/refs/heads/main/
```

```
mouseProt <- data.frame(
  y = assay(mouse[["proteins"]])["B9EKR1", ],
  celltype = mouse$celltype,
  mouse = mouse$mouse)

ggplot(mouseProt, aes(x = celltype, y = y, group = mouse, colour = mouse)) +
  geom_point() +
  geom_line() +
  theme_minimal()
```



We compare

- The two-stage analysis now summarises each mouse by the difference between its Treg and Tconv intensity (Stage I) and tests whether the average difference is zero (Stage II): a paired analysis.

- ignoring the mouse effect
- an RCB analysis with a fixed effect for mouse and
- an RCB analysis with a random intercept for mouse.

```
stageIMouse <- mouseProt |>
  pivot_wider(names_from = celltype, values_from = y) |>
  mutate(diff = reg - con)
twoStageMouse <- lm(diff ~ 1, data = stageIMouse)

wrongMouse <- lm(y ~ celltype, data = mouseProt)
rcbMouse <- lm(y ~ celltype + mouse, data = mouseProt)
lmmMouse <- lmer(y ~ celltype + (1|mouse), data = mouseProt)

rbind(
  twoStage = summary(twoStageMouse)$coef[1, 1:3],
  wrong = summary(wrongMouse)$coef[2, 1:3],
  RCB = summary(rcbMouse)$coef[2, 1:3],
  mixed = summary(lmmMouse)$coef[2, 1:3])
```

```
          Estimate Std. Error    t value
twoStage 0.4507435 0.03788201 11.8986157
wrong    0.4507435 0.70334325  0.6408585
RCB      0.4507435 0.03788201 11.8986157
mixed    0.4507435 0.03788201 11.8986157
```

- The two-stage (paired) analysis, the RCB analysis with a fixed block effect and the mixed model with a random block effect give identical results.
- Ignoring the mouse effect inflates the standard error, because the between-mouse variability then ends up in the residual variance.

The cell type effect is estimated **within** each mouse. In the difference $y_{\text{reg},i} - y_{\text{con},i} = \beta_{\text{reg}} + \epsilon_{\text{reg},i} - \epsilon_{\text{con},i}$ the mouse effect b_i cancels, so

$$\text{var}\left[\bar{y}_{\text{reg}} - \bar{y}_{\text{con}}\right] = \frac{2\sigma_\epsilon^2}{m},$$

which only involves the within-mouse variability. This is the advantage of blocking: the between-mouse variability is removed from the comparison.

Summary

Design	Treatment varies	Unit of the random effect	Variance of treatment effect	Without random effect
Pseudo-replication	between units	experimental unit (tank, bio-repeat)	between + within unit	SE too small: liberal
Blocking (RCB)	within units	block (mouse)	within unit only	SE too large: conservative

- In both designs the random intercept induces the correlation between observations of the same unit
- The mixed model automatically uses the correct variance for the treatment effect.
- In balanced designs it gives the same result as the two-stage analysis, but it also works for unbalanced designs.

Open issues:

- Mixed model is problematic if random effect variance is estimated to be zero. Then data are treated as independent
- Inference relies on asymptotic theory
- Approximate inference in small samples is possible using approximations for the degrees of freedom
- Unclear how to borrow strength across features for both random effect variance and error variance, and how to approximate the degrees of freedom in this context.

Pseudo-replication in proteomics

- Experiments with technical repeats for each biological repeat (see tutorials)
- Inference at protein-level from peptide level data

Peptide level vs protein level analysis

- Inference at protein level desired
- Observational unit: precursors/PSM level
- We perform two-stage analysis:

  1. Summarisation
  2. DA on protein summaries

- Problem: theoretically invalid due to unbalanced data due to missingness
- However, shown to work well in practice

With msqrob we started using peptide level models and explicitely addressed pseudo-replication:

$$y_{ip} = \beta_0 + \mathbf{x}_i \beta^{\text{design}} + \beta_p^{\text{peptide}} + b_i^{\text{run}} + \epsilon_{ip}$$

protein-level:

- β^{design} $\rightarrow$ DA estimates immediately estimated at the protein level from peptide level data.
- $b^{\text{run}} \sim N(0, \sigma^2_{run})$: random effect addressing pseudo-replication

peptide-level:

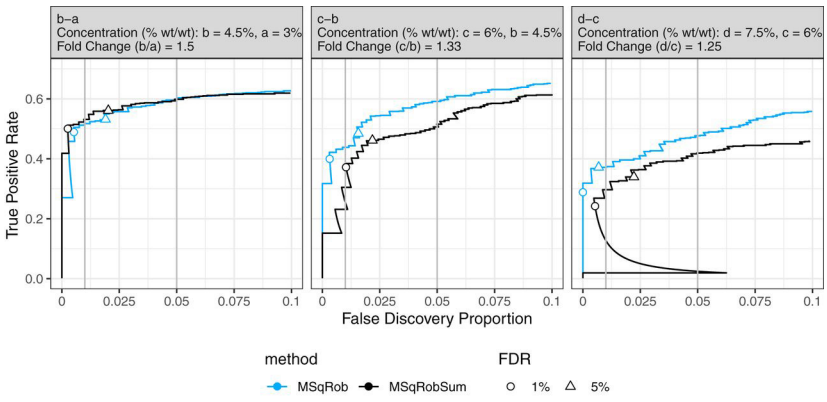
- β_p^{peptide}
- $\epsilon_{ip} \sim N(0, \sigma^2_\epsilon)$

$\rightarrow$ Still available in msqrob2 via `msqrobAggregate()`

Problems:

- Dissemination
- No protein level summaries for plotting
- Unclear which degrees of freedom to use
- For some proteins random run effect variance is set at zero: then we act as if precursor/psm/peptide level data are independent
- Computational complexity

See Sticker et al. 2020, 10.1074/mcp.RA119.001624 for comparison between peptide level and protein level models, showing that summarisation based approaches work fine when appropriate summarisation is used.

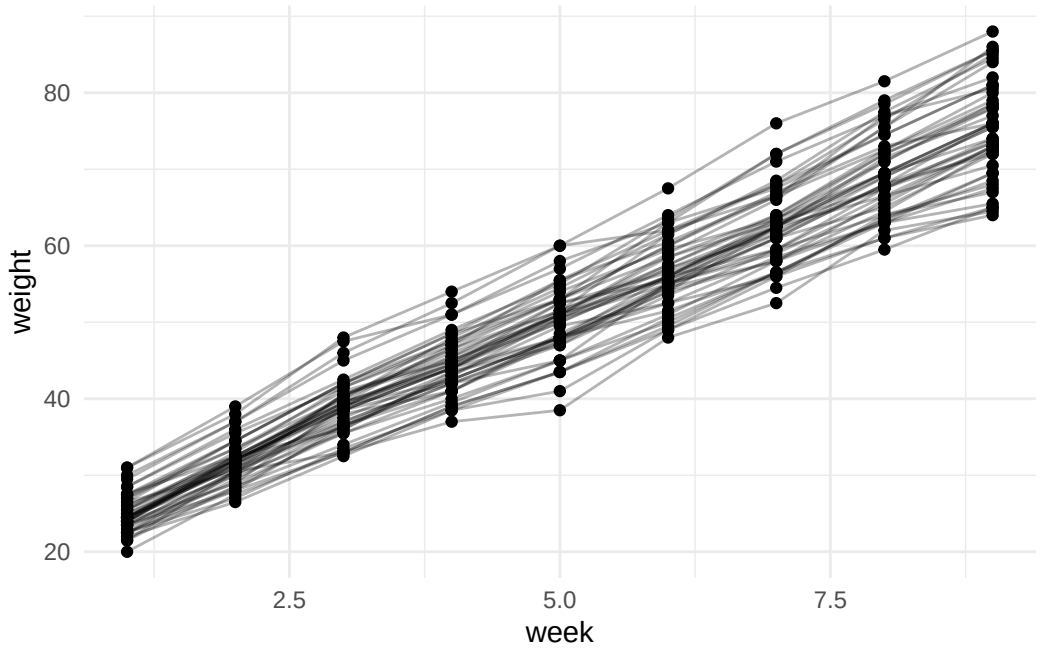


Longitudinal data: pig weights

The weight of 48 pigs was measured in 9 successive weeks (Ruppert et al. 2003). Each pig is measured repeatedly over time: the data are **longitudinal**.

```
pigs <- read.csv("data/pigweights.csv")
pigs$pig <- as.factor(pigs$pig)

ggplot(pigs, aes(x = week, y = weight, group = pig)) +
  geom_point() +
  geom_line(alpha = 0.3) +
  theme_minimal()
```



- The growth curve of an individual pig is much less variable than the cloud of all points.
- Each pig grows approximately linearly, but pigs differ in their weight at the start and in their growth rate.

We are interested in the average growth rate of pigs in the population.

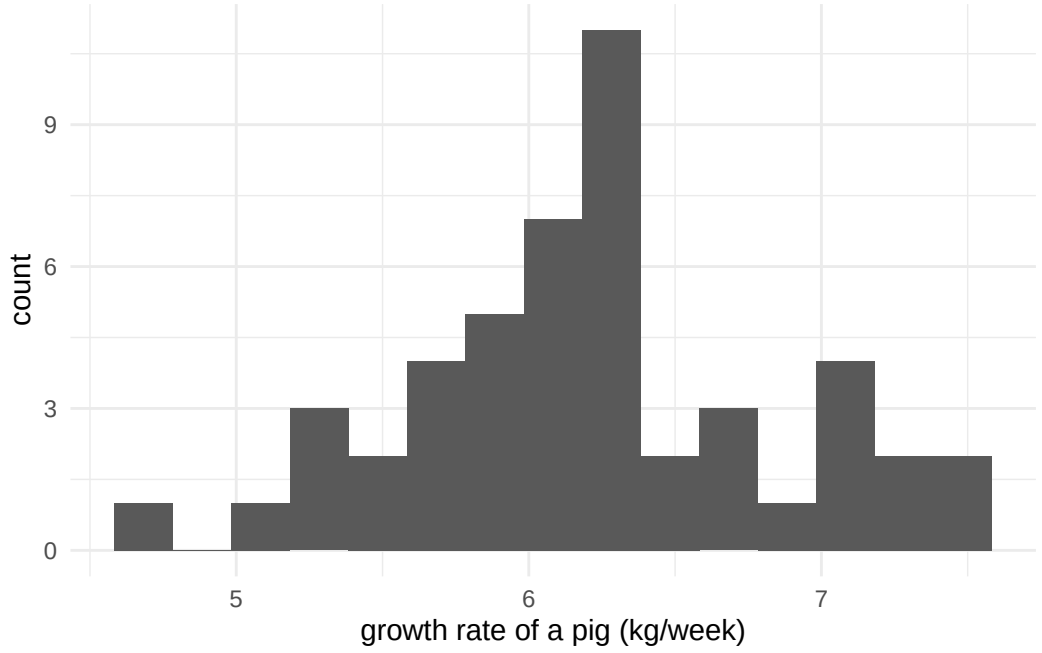
Two-stage analysis

- **Stage I**: fit a regression line for each pig i separately: $y_{ij} = \beta_{0i} + \beta_{1i}\text{week}_j + \epsilon_{ij}$.

- **Stage II**: analyse the pig-specific slopes $\hat{\beta}_{1i}$: estimate the average slope and its standard error.

```
stageIPig <- pigs |>
  group_by(pig) |>
  summarise(
    intercept = coef(lm(weight ~ week))[1],
    slope = coef(lm(weight ~ week))[2])

ggplot(stageIPig, aes(x = slope)) +
  geom_histogram(bins = 15) +
  xlab("growth rate of a pig (kg/week)") +
  theme_minimal()
```




```
twoStagePig <- lm(slope ~ 1, data = stageIPig)
summary(twoStagePig)$coef
```

```
            Estimate Std. Error  t value     Pr(>|t|)
(Intercept) 6.209896 0.09203821 67.47085 1.935218e-48
```

The standard error of the average growth rate is based on the variability of the growth rates between the 48 pigs, the experimental units. As before, the two-stage analysis loses information

and is only valid when all pigs are measured at the same time points: with drop-out, the precision of the Stage I slopes differs between pigs.

Linear mixed model with a random intercept and slope

The mixed model combines both stages:

$$y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\text{week}_j + \epsilon_{ij}, \qquad \begin{bmatrix} b_{0i} \\ b_{1i} \end{bmatrix} \sim N(\mathbf{0}, \mathbf{D}), \qquad \epsilon_{ij} \text{ i.i.d. } N(0, \sigma^2_\epsilon),$$

with

- fixed effects β_0 and β_1: the average intercept and growth rate in the population of pigs,
- random effects b_{0i} and b_{1i}: the deviation of the intercept and the growth rate of pig i from the population averages, independent between pigs,
- variance components σ^2_ϵ and $\mathbf{D} = \begin{bmatrix} \sigma^2_{b_0} & \sigma_{b_0 b_1} \\ \sigma_{b_0 b_1} & \sigma^2_{b_1} \end{bmatrix}$.

```
lmmPigSlope <- lmer(weight ~ week + (week|pig), data = pigs)
summary(lmmPigSlope)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ week + (week | pig)
   Data: pigs

REML criterion at convergence: 1740.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.6202 -0.5474  0.0150  0.5486  2.9939

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 pig      (Intercept) 6.986    2.6432
          week        0.380    0.6164   -0.06
 Residual             1.597    1.2637
Number of obs: 432, groups:  pig, 48

Fixed effects:
            Estimate Std. Error t value
```

```
(Intercept) 19.35561    0.40387   47.93
week         6.20990    0.09204   67.47

Correlation of Fixed Effects:
     (Intr)
week -0.132
```

```
rbind(
  `two-stage` = summary(twoStagePig)$coef[1, 1:2],
  `mixed model` = summary(lmmPigSlope)$coef["week", 1:2])
```

```
            Estimate Std. Error
two-stage   6.209896 0.09203821
mixed model 6.209896 0.09203775
```

The mixed model gives the same estimate and standard error of the average growth rate as the two-stage analysis, because the design is balanced: every pig is weighed in the same 9 weeks.

Why not a block analysis or a random intercept only?

We compare the mixed model with two simpler models that only account for the differences in level between the pigs:

- **Block analysis**: a fixed intercept α_i for each pig: $y_{ij} = \alpha_i + \beta_1 \text{week}_j + \epsilon_{ij}$.
- **Random intercept**: $y_{ij} = \beta_0 + b_i + \beta_1 \text{week}_j + \epsilon_{ij}$, b_i i.i.d. $N(0, \sigma_b^2)$.

```
lmFixed <- lm(weight ~ pig + week, data = pigs)
lmmPig <- lmer(weight ~ week + (1|pig), data = pigs)

rbind(
  `block analysis: fixed pig effects` = summary(lmFixed)$coef["week", 1:2],
  `random intercept` = summary(lmmPig)$coef["week", 1:2],
  `two-stage: slope per pig` = summary(twoStagePig)$coef[1, 1:2],
  `random intercept + slope` = summary(lmmPigSlope)$coef["week", 1:2])
```

```
                                Estimate Std. Error
block analysis: fixed pig effects 6.209896 0.03906329
random intercept                  6.209896 0.03906329
two-stage: slope per pig          6.209896 0.09203821
random intercept + slope          6.209896 0.09203775
```

- The block analysis and the random intercept model give a standard error that is less than half as large as the two-stage analysis: they are too liberal.
- They remove the differences in level between pigs, but assume that all pigs grow at the same rate.
- The variation of the growth rates between pigs (the pig × week interaction) then ends up in the residuals.
- This is again a form of pseudo-replication: the 9 measurements of a pig are treated as 9 independent pieces of information on the growth rate in the population.

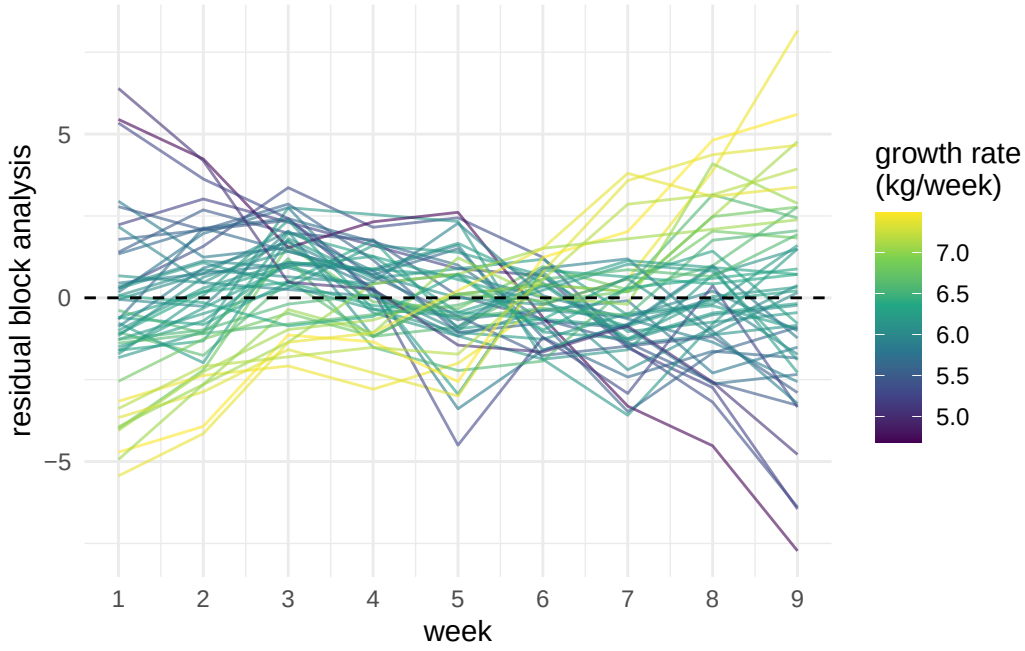
Whenever an effect is assessed repeatedly within an experimental unit, its variation between the units has to be part of its standard error!

Residuals of the block analysis

We colour the pigs according to their growth rate from Stage I of the two-stage analysis.

```
pigs$resFixed <- residuals(lmFixed)
pigs <- pigs |>
  left_join(stageIPig |> dplyr::select(pig, slope), by = "pig")

ggplot(pigs, aes(x = week, y = resFixed, group = pig, colour = slope)) +
  geom_line(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_colour_viridis_c(name = "growth rate\n(kg/week)") +
  scale_x_continuous(breaks = 1:9) +
  ylab("residual block analysis") +
  theme_minimal()
```



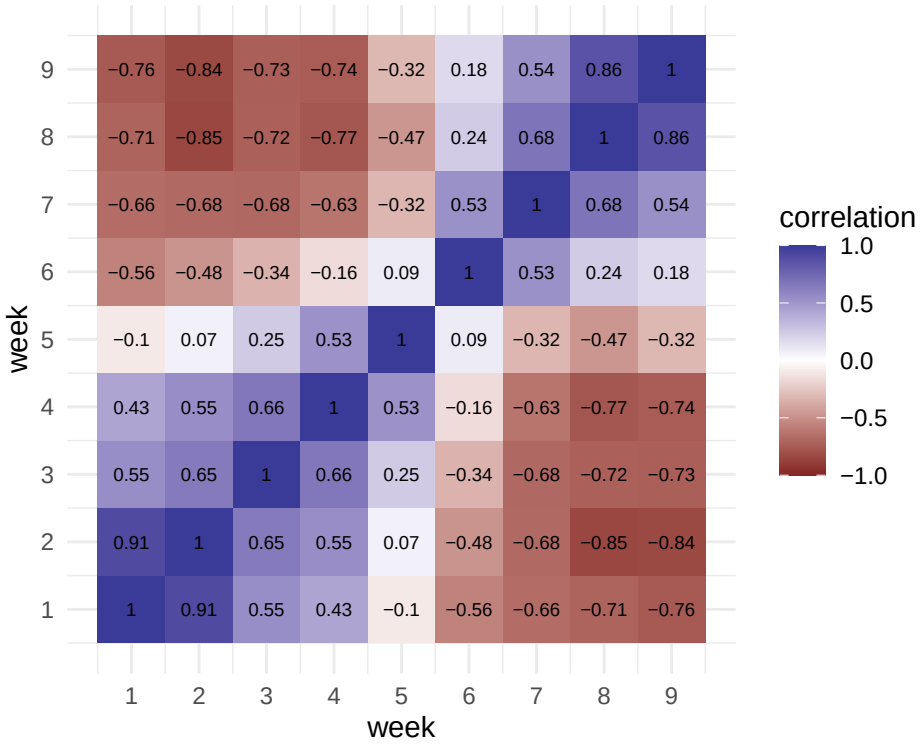
- The fixed pig effects remove the differences in level between the pigs, but the residuals of a pig are not random noise around zero.
- Pigs that grow faster than average have negative residuals in the early weeks and positive residuals in the late weeks, and vice versa for pigs that grow slowly: the pig × week interaction ends up in the residuals.

```
# correlation heatmap of a 9 x 9 correlation matrix
corHeatmap <- function(corMat) {
  tibble(
    week1 = rep(1:9, times = 9),
    week2 = rep(1:9, each = 9),
    correlation = as.vector(corMat))
}

resFixedWide <- pigs |>
  dplyr::select(pig, week, resFixed) |>
  pivot_wider(names_from = week, values_from = resFixed)

corHeatmap(cor(resFixedWide[, -1])) |>
  ggplot(aes(x = week1, y = week2, fill = correlation)) +
  geom_tile() +
  geom_text(aes(label = round(correlation, 2)), size = 2.5) +
  scale_fill_gradient2(limits = c(-1, 1)) +
```

```
  scale_x_continuous(breaks = 1:9) +
  scale_y_continuous(breaks = 1:9) +
  coord_equal() +
  xlab("week") +
  ylab("week") +
  theme_minimal()
```



- Residuals of the same pig in nearby weeks are positively correlated, residuals in early and late weeks are negatively correlated: the independence assumption of the block analysis is violated.
- The within-pig error is not underestimated by the block analysis: its residual standard deviation, 2.1, is even larger than that of the random slope model, 1.26.
- The problem is that it uses the wrong error term to assess the growth rate.

We see this by splitting its residual sum of squares into two parts by adding the pig × week interaction, i.e. a separate slope for each pig:

```
lmPigSlopes <- lm(weight ~ pig + week + pig:week, data = pigs)
ssTab <- anova(lmFixed, lmPigSlopes)

decomp <- tibble(
  source = c(
    "differences in slope between pigs (pig x week)",
    "scatter around the line of each pig",
    "residual block analysis (pooled)"),
  SS = c(ssTab$`Sum of Sq`[2], ssTab$RSS[2], ssTab$RSS[1]),
  df = c(ssTab$Df[2], ssTab$Res.Df[2], ssTab$Res.Df[1])) |>
  mutate(MS = SS / df)
decomp
```

```
# A tibble: 3 x 4
  source                                            SS    df    MS
  <chr>                                          <dbl> <dbl> <dbl>
1 differences in slope between pigs (pig x week) 1147.    47 24.4
2 scatter around the line of each pig             537.   336  1.60
3 residual block analysis (pooled)               1683.   383  4.39
```

- The growth rate is assessed within each pig, so its standard error has to be based on how much the growth rates vary between the pigs: the first row, with 47 degrees of freedom.
- The block analysis pools the slope differences with the scatter around the pig-specific lines and divides by 383 instead of 47 degrees of freedom: the variability of the growth rates is diluted and the error term becomes much too small.

With $n = 48$ pigs and $S_{xx} = \sum_{j=1}^{9}(\text{week}_j - \overline{\text{week}})^2 = 60$, the standard error of the average growth rate is $\sqrt{\text{MS}/(nS_{xx})}$:

```
Sxx <- sum((1:9 - mean(1:9))^2)
nPig <- nlevels(pigs$pig)
c(`MS pig x week (correct)` = sqrt(decomp$MS[1] / (nPig * Sxx)),
  `MS pooled (block analysis)` = sqrt(decomp$MS[3] / (nPig * Sxx)))
```

```
   MS pig x week (correct) MS pooled (block analysis)
                0.09203821                 0.03906329
```

- With the mean square of the pig × week interaction we recover the standard error of the two-stage analysis and of the random slope model.
- With the pooled mean square the much too small standard error of the block analysis.
- As for the fish tanks: an effect has to be assessed against the variability of the units at which it varies, here the variability of the growth rates between pigs.

Correlation between measurements of the same pig

The random intercept model implies **compound symmetry**: all measurements of the same pig have the same variance $\sigma_b^2 + \sigma_\epsilon^2$ and the same correlation $\sigma_b^2/(\sigma_b^2 + \sigma_\epsilon^2)$, whatever the time between them. That is natural for fish in a tank or MS-runs of a bio-repeat, but not for repeated measurements over time when subjects grow at different rates.

With a random intercept and slope, the variance and the covariance depend on the weeks at which the measurements were taken:

$$\text{cov}\left[y_{ij}, y_{ij'}\right] = \sigma_{b_0}^2 + \sigma_{b_0 b_1}(\text{week}_j + \text{week}_{j'}) + \sigma_{b_1}^2 \text{week}_j \text{week}_{j'} + \sigma_\epsilon^2 \delta_{jj'},$$

with $\delta_{jj'} = 1$ if $j = j'$ and 0 otherwise.

We compare the observed correlation between the weights of the same pig in different weeks with the correlations implied by both models.

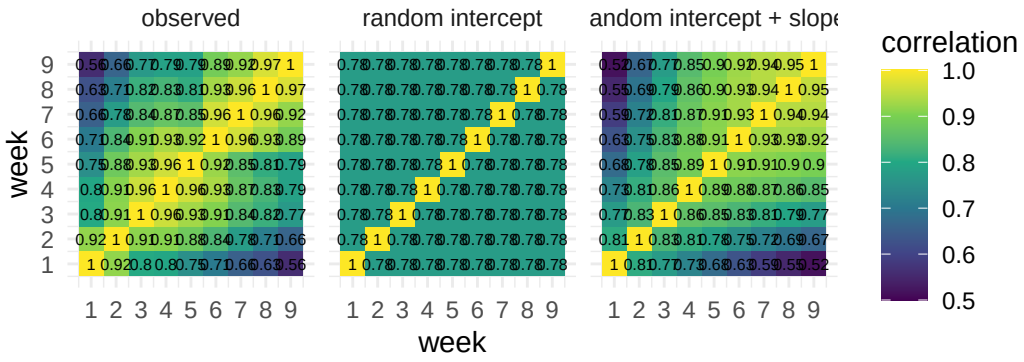
```
# implied correlation matrix of the 9 measurements of a pig
impliedCor <- function(fit, Zi) {
  D <- bdiag(VarCorr(fit)) |> as.matrix()
  Vi <- Zi %*% D %*% t(Zi) + diag(nrow(Zi)) * sigma(fit)^2
  cov2cor(Vi)
}

weightWide <- pigs |>
  dplyr::select(pig, week, weight) |>
  pivot_wider(names_from = week, values_from = weight)

corMats <- list(
  observed = cor(weightWide[, -1]),
  `random intercept` = impliedCor(lmmPig, matrix(1, 9, 1)),
  `random intercept + slope` = impliedCor(lmmPigSlope, cbind(1, 1:9)))

corMats |>
  map(corHeatmap) |>
  bind_rows(.id = "matrix") |>
  mutate(matrix = factor(matrix, levels = names(corMats))) |>
  ggplot(aes(x = week1, y = week2, fill = correlation)) +
  geom_tile() +
  geom_text(aes(label = round(correlation, 2)), size = 2) +
  scale_fill_viridis_c(limits = c(0.5, 1)) +
  scale_x_continuous(breaks = 1:9) +
  scale_y_continuous(breaks = 1:9) +
```

```
  coord_equal() +
  facet_wrap(~ matrix) +
  xlab("week") +
  ylab("week") +
  theme_minimal()
```



- The random intercept model implies the same correlation for all pairs of weeks.
- In the observed data the correlation decreases with the time between the measurements, and the random intercept and slope model captures this pattern.
- In the early weeks the weight of a pig is mainly determined by its random intercept, in the later weeks by its random slope, whose contribution grows with time. Hence, measurements in the later weeks are more strongly correlated than measurements in the early weeks.

Random treatment effects in block designs

The random slope for the pigs is a random treatment effect: the effect of the "treatment" (time) varies from block (pig) to block. Do we always need such a random block × treatment effect when we have repeated measures within a block? No: we need it when the residual no longer contains the variability of the treatment effect between blocks. This happens as soon as there is more than one observation per block × treatment combination.

Design	Example	Model	Random block × treatment effect
One observation per block × treatment	mouse T-cells: one Treg and one Tconv sample per mouse	y ~ treatment + (1 | block)	not needed: confounded with ϵ
Technical repeats of each block × treatment	Treg and Tconv sample of each mouse run in technical triplicate	y ~ treatment + (1 | block) + (1 | block:treatment)	always needed: block × treatment is the experimental unit
Several experimental units per block × treatment	several mice per litter per treatment	y ~ treatment + (1 | block) + (1 | block:treatment)	needed when the treatment effect varies between blocks
Continuous treatment within block	pig weights over time	y ~ week + (week | pig)	random slope, needed when slopes vary between pigs

- **Technical repeats**: omitting the block × treatment effect is pseudo-replication, as in the fish tanks. In a balanced design, the mixed model is equivalent to averaging the technical repeats per block × treatment and an RCB analysis on these averages.
- **Several experimental units per block × treatment**: the random effect can be omitted when its variance is zero (Section), but with few blocks keeping it is the safe default.
- **Peptide-level models**: the random run effect b_i^{run} plays the role of the block × treatment effect: the peptides are repeated measures within each run.

Note that a standard `lm` analysis on the observational units is never correct in these designs (unless the block × treatment variance is zero): it always tests against the residual mean square. In balanced designs, a two-stage analysis on the block × treatment averages, or on the block-specific slopes, gives the correct standard error.

Prediction of random effects

The fixed effects and the variance components are estimated, but the random effects are **predicted**: the target is random, not a fixed parameter.

Best prediction

Suppose we observe $y = v + \epsilon$ with $v \sim N(0, 1)$ and $\epsilon \sim N(0, 4)$ independent. What is the best guess for the unobserved v?

The **best predictor** minimises $E\left[(\tilde{v}-v)^2\right]$ and equals $\tilde{v}=E\left[v|y\right]$. Because v and y are jointly normal, $E\left[v|y\right]$ is the regression line of v on y:

$$E\left[v|y\right]=E\left[v\right]+\frac{\text{cov}\left[v,y\right]}{\text{var}\left[y\right]}\{y-E\left[y\right]\}=\frac{1}{5}y,$$

because $\text{cov}\left[v,y\right]=\text{var}\left[v\right]=1$, $\text{var}\left[y\right]=\text{var}\left[v\right]+\text{var}\left[\epsilon\right]=5$ and $E\left[v\right]=E\left[y\right]=0$. Only 1/5 of the variability in y comes from v, so we only keep 1/5 of what we observe: the prediction is a *shrunken* version of y.

Equivalently, $\tilde{v}$ is a **precision-weighted average** of the prior mean $E\left[v\right]=0$, with precision 1, and the observation y, with precision 1/4:

$$\tilde{v}=\frac{1\times 0+\frac{1}{4}\times y}{1+\frac{1}{4}}=\frac{y}{5}.$$

The noisier the observation, the stronger the shrinkage towards the mean of v.

Fish tanks

For the random intercept model of the fish, the best predictor of the effect of tank t is

$$\tilde{b}_t=\frac{n\sigma_b^2}{\sigma_\epsilon^2+n\sigma_b^2}(\bar{y}_{t\cdot}-\beta_0-\beta_{d(t)}),$$

i.e. the deviation of the tank average from its diet mean, shrunken towards 0.

This is again a precision-weighted average, as for $\tilde{v}=y/5$. We have two sources of information on b_t:

- the prior mean $E\left[b_t\right]=0$, with precision $1/\text{var}\left[b_t\right]=1/\sigma_b^2$,
- the observed deviation $\bar{y}_{t\cdot}-\beta_0-\beta_{d(t)}=b_t+\bar{\epsilon}_{t\cdot}$, with precision $1/\text{var}\left[\bar{\epsilon}_{t\cdot}\right]=n/\sigma_\epsilon^2$: the average of n fish.

Hence

$$\tilde{b}_t=\frac{\frac{1}{\sigma_b^2}\times 0+\frac{n}{\sigma_\epsilon^2}(\bar{y}_{t\cdot}-\beta_0-\beta_{d(t)})}{\frac{1}{\sigma_b^2}+\frac{n}{\sigma_\epsilon^2}}=\frac{n\sigma_b^2}{\sigma_\epsilon^2+n\sigma_b^2}(\bar{y}_{t\cdot}-\beta_0-\beta_{d(t)}).$$

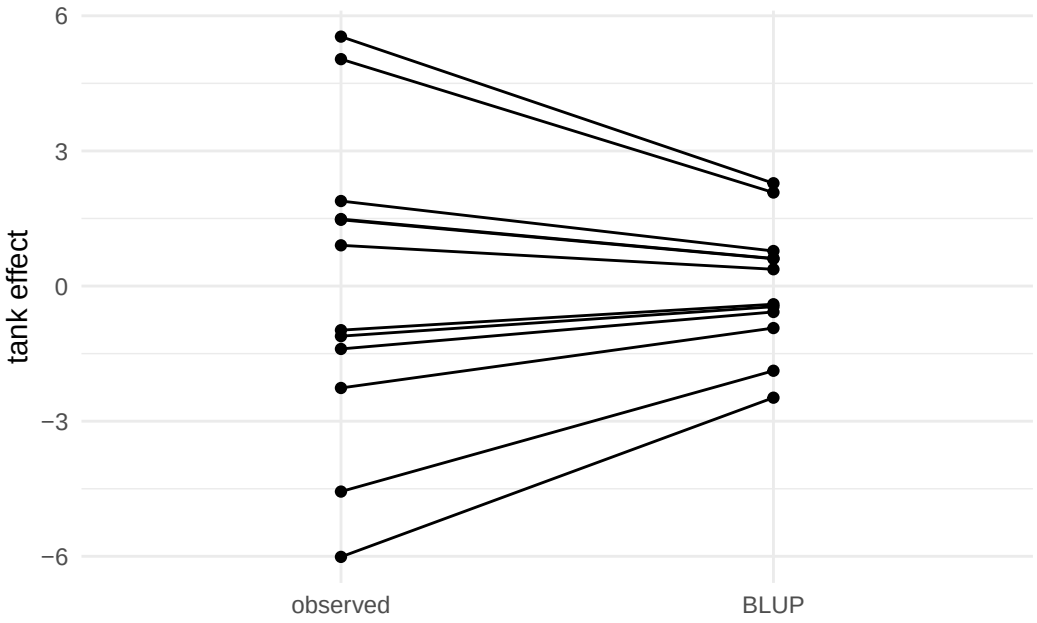
The shrinkage is strong when there are few observations per unit or when σ_ϵ^2 is large compared to σ_b^2: then the tank average is imprecise compared to the prior information that tank effects are close to 0.

```
vcFish <- as.data.frame(VarCorr(lmmFish))
sigma2b <- vcFish$vcov[vcFish$grp == "tank"]
sigma2EpsFish <- vcFish$vcov[vcFish$grp == "Residual"]
nFish <- 6

shrinkFish <- nFish * sigma2b / (sigma2EpsFish + nFish * sigma2b)
shrinkFish
```

```
[1] 0.4121706
```

```
dietMeans <- predict(lmmFish, newdata = stageIFish, re.form = NA)
tibble(
  tank = stageIFish$tank,
  observed = stageIFish$wtGain - dietMeans,
  BLUP = ranef(lmmFish)$tank[, 1]) |>
  pivot_longer(-tank) |>
  mutate(name = factor(name, levels = c("observed", "BLUP"))) |>
  ggplot(aes(x = name, y = value, group = tank)) +
  geom_point() +
  geom_line() +
  xlab("") +
  ylab("tank effect") +
  theme_minimal()
```



With only 6 fish per tank and a within-tank variance that is much larger than the between-tank variance, the tank effects are shrunken strongly towards zero.

The linear mixed model in general

The theory follows Ruppert et al. (2003), Chapter 4. In general, the linear mixed model (LMM) is

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{b} + \epsilon, \qquad \mathbf{b} \sim N(\mathbf{0}, \mathbf{G}), \qquad \epsilon \sim N(\mathbf{0}, \mathbf{R}),$$

with $\mathbf{b}$ and ϵ independent. For the pigs, $\mathbf{X}$ has a column of ones and a column with the week, $\mathbf{b}$ contains the 48 random intercepts and the 48 random slopes, $\mathbf{Z}$ contains the pig indicators and the pig indicators multiplied by the week, and $\mathbf{R} = \sigma^2_\epsilon \mathbf{I}$.

The LMM can be viewed in two ways.

- **Conditional model**: given the random effects, the data follow an ordinary linear model, $\mathbf{y}|\mathbf{b} \sim N(\mathbf{X}\beta + \mathbf{Z}\mathbf{b}, \mathbf{R})$: the weights of a pig vary independently around the pig-specific growth line.
- **Marginal model**: if we do not condition on the random effects, the data are correlated,

$$\mathbf{y} \sim N(\mathbf{X}\beta, \mathbf{V}), \qquad \mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^T + \mathbf{R}:$$

weights of the same pig are correlated because they share the same random effects, and weights of different pigs are independent.

Estimation and prediction

Likelihood

The parameters of the LMM are estimated by maximum likelihood.

- We start from the observed sample and consider it as fixed and known: the observations $\mathbf{y}_1, \ldots, \mathbf{y}_m$ of the m experimental units, e.g. the vectors with the 9 weights of each of the 48 pigs.
- We view them as a realisation of random vectors $\mathbf{Y}_1, \ldots, \mathbf{Y}_m$ with a density $f(\mathbf{y}_i; \theta)$ that depends on unknown parameters θ, e.g. a multivariate normal distribution $N(\mathbf{X}_i\beta, \mathbf{V}_i)$ with θ the fixed effects β and the variance components in $\mathbf{V}_i$.
- The **likelihood** measures how probable it is to observe this sample for a given value of the parameters. When the experimental units are independent, it is the product of the densities of the units, viewed as a function of the parameters:

$$L(\theta) = \prod_{i=1}^{m} f(\mathbf{y}_i; \theta).$$

- In practice we use the **log-likelihood**, which turns the product into a sum:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{m} \log f(\mathbf{y}_i; \theta).$$

- **Maximum likelihood** (ML) estimates the parameters by the values that maximise $\ell(\theta)$: the parameter values under which the observed sample is most probable.

Linear model. For $\mathbf{y} = \mathbf{X}\beta + \epsilon$ with $\epsilon \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, the log-likelihood is

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta).$$

- Setting the derivative with respect to β to zero gives $\mathbf{X}^T(\mathbf{y} - \mathbf{X}\widehat{\beta}) = \mathbf{0}$, i.e. the least squares estimator $\widehat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.
- Setting the derivative with respect to σ^2 to zero gives $\widehat{\sigma}^2 = (\mathbf{y} - \mathbf{X}\widehat{\beta})^T(\mathbf{y} - \mathbf{X}\widehat{\beta})/n$, which is biased: it divides by n instead of $n - p$.

For the LMM we can write down two likelihoods, one for each view of the model.

LMM - Marginal likelihood. The marginal model $\mathbf{y} \sim N(\mathbf{X}\beta, \mathbf{V})$ gives

$$\ell(\beta, \mathbf{G}, \mathbf{R}) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{V}| - \frac{1}{2}(\mathbf{y}-\mathbf{X}\beta)^T\mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\beta), \qquad \mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^T + \mathbf{R}.$$

It is used to estimate the fixed effects and the variance components.

- Setting the derivative with respect to β to zero gives $\mathbf{X}^T\mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\hat{\beta}) = \mathbf{0}$: the generalised least squares estimator $\hat{\beta} = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{V}^{-1}\mathbf{y}$, with var $\left[\hat{\beta}\right] = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}$.
- There is no closed-form solution for the variance components in $\mathbf{G}$ and $\mathbf{R}$: the likelihood is maximised numerically.
- As for the linear model, ML underestimates the variance components because it ignores the degrees of freedom used to estimate β. REML corrects for this by maximising the restricted log-likelihood

$$\ell_R = \ell(\hat{\beta}, \mathbf{G}, \mathbf{R}) - \frac{1}{2}\log|\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X}|.$$

LMM -Joint likelihood: Henderson's method. The hierarchical model $\mathbf{y}|\mathbf{b} \sim N(\mathbf{X}\beta + \mathbf{Z}\mathbf{b}, \mathbf{R})$ and $\mathbf{b} \sim N(\mathbf{0}, \mathbf{G})$ give the joint log-likelihood of the data and the random effects:

$$\begin{aligned} \ell(\beta, \mathbf{b}) &= \log f(\mathbf{y}|\mathbf{b}) + \log f(\mathbf{b}) \\ &= \text{constant} - \tfrac{1}{2}\left\{(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\mathbf{b})^T\mathbf{R}^{-1}(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\mathbf{b}) + \mathbf{b}^T\mathbf{G}^{-1}\mathbf{b}\right\}. \end{aligned}$$

Henderson proposed to maximise this joint likelihood over β and $\mathbf{b}$ simultaneously, for given variance components. This amounts to minimising

$$(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\mathbf{b})^T\mathbf{R}^{-1}(\mathbf{y}-\mathbf{X}\beta-\mathbf{Z}\mathbf{b}) + \mathbf{b}^T\mathbf{G}^{-1}\mathbf{b}.$$

- The first term is the (generalised) least squares criterion of a linear model in which the random effects $\mathbf{b}$ are treated as fixed effects.
- The second term penalises large random effects.
- Hence, the BLUP is **generalised least squares with a ridge penalty on the random effects only**: the fixed effects are not penalised.

With $\mathbf{C} = [\mathbf{X}\ \mathbf{Z}]$ and the penalty matrix $\mathbf{B} = \text{blockdiag}(\mathbf{0}, \mathbf{G}^{-1})$, the solution is

$$\begin{bmatrix} \tilde{\beta} \\ \tilde{\mathbf{b}} \end{bmatrix} = (\mathbf{C}^T\mathbf{R}^{-1}\mathbf{C} + \mathbf{B})^{-1}\mathbf{C}^T\mathbf{R}^{-1}\mathbf{y}.$$

Henderson showed that this gives exactly the GLS estimator $\tilde{\beta}$ of the marginal likelihood and the BLUP $\tilde{\mathbf{b}} = \mathbf{G}\mathbf{Z}^T\mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\tilde{\beta})$. Without the penalty $\mathbf{B}$, it would be the least squares fit of the model with fixed unit effects.

Setting the derivative of Henderson's criterion with respect to $\mathbf{b}$ to zero shows where the shrinkage comes from. With $\mathbf{R} = \sigma_\epsilon^2 \mathbf{I}$:

$$\tilde{\mathbf{b}} = (\mathbf{Z}^T\mathbf{Z} + \sigma_\epsilon^2 \mathbf{G}^{-1})^{-1}\mathbf{Z}^T(\mathbf{y} - \mathbf{X}\tilde{\beta}).$$

- Without the term $\sigma_\epsilon^2 \mathbf{G}^{-1}$ this is the least squares solution $(\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T(\mathbf{y} - \mathbf{X}\tilde{\beta})$: for the fish, the tank averages minus the diet means; for the pigs, the Stage I regression lines minus the population line. The term $\sigma_\epsilon^2 \mathbf{G}^{-1}$ shrinks the random effects towards zero.
- For the fish tanks, $\mathbf{Z}^T\mathbf{Z} = n\mathbf{I}$ and $\mathbf{G} = \sigma_b^2\mathbf{I}$, so $\tilde{b}_t = \frac{n}{n+\sigma_\epsilon^2/\sigma_b^2}(\bar{y}_{t\cdot} - \beta_0 - \beta_{d(t)})$: exactly the shrinkage factor $n\sigma_b^2/(n\sigma_b^2 + \sigma_\epsilon^2)$ of the fish tanks.

Estimators in practice

- **Fixed effects**: generalised least squares, which accounts for the correlation in $\mathbf{V}$: $\tilde{\beta} = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{V}^{-1}\mathbf{y}$.
- **Random effects**: the best linear unbiased predictor (BLUP), $\tilde{\mathbf{b}} = E[\mathbf{b}|\mathbf{y}] = \mathbf{G}\mathbf{Z}^T\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\tilde{\beta})$. As for the fish tanks, these are the unit-specific estimates shrunken towards zero.
- **Variance components** in $\mathbf{G}$ and $\mathbf{R}$: maximum likelihood (ML) or restricted maximum likelihood (REML). REML maximises the likelihood of the residuals after fitting the fixed effects, whose distribution does not depend on β. It therefore accounts for the degrees of freedom used to estimate the fixed effects, just like the sample variance divides by $n-1$ instead of n. REML is preferred in small samples and is the default in `lmer`.

In practice the estimated variance components are plugged in: `fixef()` and `ranef()` return these estimated BLUPs (EBLUPs).

Inference

- The standard errors of the fixed effects follow from $\text{var}\left[\tilde{\beta}\right] = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}$ with $\hat{\mathbf{V}}$ plugged in. This ignores the uncertainty on the variance components, which matters in small experiments. Small-sample corrections approximate the degrees of freedom of the t-statistics, e.g. Satterthwaite or Kenward-Roger (packages `lmerTest` and `pbkrtest`).
- Fixed effects can also be tested with a likelihood ratio test (LRT). Use ML, not REML, to compare models with different fixed effects: the restricted likelihood is the likelihood of the residuals after fitting the fixed effects, so it is not comparable across different fixed effects models.

```
lmmPigSlopeML <- lmer(weight ~ week + (week|pig), data = pigs, REML = FALSE)
lmmPigSlopeML0 <- lmer(weight ~ 1 + (week|pig), data = pigs, REML = FALSE)
anova(lmmPigSlopeML0, lmmPigSlopeML)
```

```
Data: pigs
Models:
lmmPigSlopeML0: weight ~ 1 + (week | pig)
lmmPigSlopeML: weight ~ week + (week | pig)
               npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
lmmPigSlopeML0    5 1967.9 1988.3 -978.97    1957.9
lmmPigSlopeML     6 1749.9 1774.3 -868.96    1737.9 220.01  1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Testing a variance component

Under $H_0 : \sigma_b^2 = 0$ the variance is on the **boundary** of its parameter space $[0, \infty)$, so the classical χ^2 approximation of the LRT does not hold. For a random intercept, the LRT follows a 50:50 mixture of a point mass at zero and a χ^2_1 distribution (Self and Liang 1987): in about half of the datasets $\hat{\sigma}_b^2 = 0$. Using χ^2_1 gives p-values that are twice too large.

To test whether we need a random slope, the LRT follows a 50:50 mixture of a χ^2_1 and a χ^2_2 distribution (the covariance between random intercept and slope also vanishes under H_0). Both models have the same fixed effects, so we can use their restricted likelihoods.

```
lrtSlope <- 2 * (as.numeric(logLik(lmmPigSlope)) - as.numeric(logLik(lmmPig)))
c(LRT = lrtSlope,
  pval = 0.5 * pchisq(lrtSlope, df = 1, lower.tail = FALSE) +
    0.5 * pchisq(lrtSlope, df = 2, lower.tail = FALSE))
```

```
         LRT         pval
2.929256e+02 1.290343e-64
```

The random slope is clearly needed. With few subjects the mixture approximations are inaccurate and the null distribution is best obtained by simulation (parametric bootstrap).

Summary

Experimental unit	unit to which the treatment is randomised; its variability has to be part of the standard error of the treatment effect

Two-stage analysis	summarise each experimental unit (Stage I), analyse the summaries (Stage II); only valid for balanced designs
Pseudo-replication	treatment varies between units: random effect for the experimental unit
Blocking	treatment varies within units: random or fixed block effect, and a random block × treatment effect if there are repeats within block × treatment
Model	$\mathbf{y} = \mathbf{X}\beta + \mathbf{Zb} + \epsilon$, $\text{var}[\mathbf{y}] = \mathbf{V} = \mathbf{ZGZ}^T + \mathbf{R}$
Estimation	GLS for β, BLUP (shrinkage) for $\mathbf{b}$, REML for the variance components
Inference	approximate degrees of freedom; LRT with ML for fixed effects; boundary problem for variance components

Ruppert, D., M. P. Wand, and R. J. Carroll. 2003. *Semiparametric Regression*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press.

Self, Steven G., and Kung-Yee Liang. 1987. "Asymptotic Properties of Maximum Likelihood Estimators and Likelihood Ratio Tests Under Nonstandard Conditions." *Journal of the American Statistical Association* 82 (398): 605–10.